Generalized Euler-Lagrange equation.

Integral Master

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We have M a manifold (possibly spacetime) with $\Omega \subset M$ and $u_i : \Omega \to M$ a collection of fields with $i \in \mathbb{N} \bigcap [0, m], m \in \mathbb{N}$, we write $\mathcal{L} : M \times TM \to \mathbb{R}$ as the Lagrangian, explicitly $\mathcal{L}(u^i, \nabla \cdot u^i)$ where $u^i = (u_1, \ldots, u_m)$, also, we use the mathematical convention for ∇ in:

$$\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n} \right)$$

and we will translate it to physical systems accordingly.

$$S[u^i] = \int_\Omega \mathcal{L}(u^i, \nabla \cdot u^i) dV$$

$$\delta S = \lim_{\varepsilon \to 0} \frac{S[u^i + \varepsilon \eta^i] - S[u^i]}{\varepsilon} = \frac{d}{d\varepsilon} S[u^i + \varepsilon \eta^i] \Big|_{\varepsilon = 0}$$

. Choose $\eta_k \in \mathcal{C}_c^\infty(\Omega)$ such that $\eta_k \left(\partial \Omega \right) = \{0\}$ and let $(\forall j \neq k) : \eta_j = 0$.

$$\delta S = \frac{d}{d\varepsilon} \int_\Omega \mathcal{L}(u^i + \varepsilon \eta^i, \nabla \cdot u^i + \varepsilon \nabla \cdot \eta^i) dV \Big|_{\varepsilon = 0}$$

$$= \int_\Omega \sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial (u_i + \varepsilon \eta_i)} \frac{d}{d\varepsilon} \left(u_i + \varepsilon \eta_i \right) + \frac{\partial \mathcal{L}}{\partial (\nabla u_i + \varepsilon \nabla \eta_i)} \cdot \frac{d}{d\varepsilon} \left(\nabla u_i + \varepsilon \nabla \eta_i \right) dV \Big|_{\varepsilon = 0}$$

$$= \int_\Omega \sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial (u_i + \varepsilon \eta_i)} \eta_i + \frac{\partial \mathcal{L}}{\partial (\nabla u_i + \varepsilon \nabla \eta_i)} \cdot \nabla \eta_i dV \Big|_{\varepsilon = 0} = \int_\Omega \sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial u_i} \eta_i + \frac{\partial \mathcal{L}}{\partial (\nabla u_i)} \cdot \nabla \eta_i dV.$$

Since $\eta_j = 0, \forall j \neq k$:

$$\int_\Omega \sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial u_i} \eta_i + \frac{\partial \mathcal{L}}{\partial (\nabla u_i)} \cdot \nabla \eta_i dV = \int_\Omega \frac{\partial \mathcal{L}}{\partial u_k} \eta_k + \frac{\partial \mathcal{L}}{\partial (\nabla u_k)} \cdot \nabla \eta_k dV.$$

The product rule for gradients states:

$$\nabla \cdot (\mathbf{A} f) = (\nabla \cdot \mathbf{A}) f + \mathbf{A} \cdot \nabla f$$

Put:

$$\mathbf{A} = \frac{\partial \mathcal{L}}{\partial (\nabla u_k)}, f = \eta_k$$

$$\nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \eta_k \right) = \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) \eta_k + \frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \cdot \nabla \eta_k$$

$$\implies \frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \cdot \nabla \eta_k = \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \eta_k \right) - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) \eta_k$$

$$\implies \int_\Omega \frac{\partial \mathcal{L}}{\partial u_k} \eta_k + \frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \cdot \nabla \eta_k dV = \int_\Omega \frac{\partial \mathcal{L}}{\partial u_k} \eta_k + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \eta_k \right) - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) \eta_k dV$$

$$= \int_\Omega \left(\frac{\partial \mathcal{L}}{\partial u_k} - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) \right) \eta_k dV + \int_\Omega \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \eta_k \right) dV.$$

By the divergence theorem, with $\hat{n}$ being the unit normal vector to $\partial\Omega$

$$\int_\Omega \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \eta_k \right) dV = \int_{\partial\Omega} \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \eta_k \right) \cdot \hat{n} dS = 0$$

since $\eta_k(\partial\Omega) = \{0\}$, therefore:

$$\delta S = \int_\Omega \left(\frac{\partial \mathcal{L}}{\partial u_k} - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) \right) \eta_k dV.$$

Invoking Hamilton's principle:

$$\delta S = 0 \implies \int_\Omega \left(\frac{\partial \mathcal{L}}{\partial u_k} - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) \right) \eta_k dV = 0.$$

By the fundamental lemma of the Calculus of Variations:

$$\frac{\partial \mathcal{L}}{\partial u_k} - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right) = 0 \iff \frac{\partial \mathcal{L}}{\partial u_k} = \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u_k)} \right)$$